

Younsoo Park

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Education

The Pennsylvania State University

University Park, PA

Bachelor of Science in Computer Science Minor in Mathematics Expected Graduation: May 2027 Dean's List

Research Interests

Systems Security
Network Security
Distributed Systems
LLM Systems
AI Agent Memory
Federated Learning
TinyML for IoT Security
Adversarial Machine Learning
Edge AI Systems

Research Experience

Undergraduate Research Assistant & Team Lead

Oct 2025 – Present

Reliable Federated TinyML Deployment for IoT Security (WIP, LCTES '26)

Penn State University and Elizabethtown College Supervisors: Dr. Suman Saha and Dr. Peilong Li

Research focus: Federated and adversarially robust TinyML systems for IoT security and resource-constrained edge-device deployment.

Key contributions:

- Designed a federated learning pipeline using FedAvgM, focal loss, and cosine learning-rate scheduling on CIC-IDS2017.
- Improved attack recall from 46.7% to 93.85% and F1-score from 84.1% to 89.32%.
- Built a TinyML compression pipeline including BatchNorm folding, structured pruning, knowledge distillation, and INT8 QAT/PTQ.
- Achieved $12.28\times$ model compression and 74.5% inference latency reduction for ESP32 deployment.
- Evaluated adversarial robustness using PGD attacks (ϵ -sweep 0.01–0.2) across 48 compression configurations.

Publications and Manuscripts

Reliable Federated TinyML Deployment for IoT Security Work in Progress Target Venue: ACM LCTES 2026

Independent Study

AI Agent Memory Systems

Jul 2026 – Present

Advisor: Prof. Kiwan Maeng

- Researched five production and research AI agent memory systems (Mem0, A-MEM, LangMem, Zep, OpenAI Memory) to identify the SOTA approach, comparing memory creation, retrieval, and lifecycle-management approaches.
- Identified a core architectural divergence: Mem0 extracts and retrieves discrete facts, while A-MEM models memory as an evolving, interconnected knowledge structure; traced how Mem0's open-source implementation diverged from its published ADD/UPDATE/DELETE design into an ADD-only, retrieval-time resolution approach.
- Narrowed research focus to Mem0 (implementation depth, practical extensibility) and A-MEM (architectural novelty) for application to the lab's production AI agent system.
- Applying memory architectures to production recommendation systems through a Memory-first Personalization framework, shipped as Wowness at Levit Inc.

Industry and Technical Experience

Levit Inc. (Shoppport AI)

Seoul, South Korea

Associate Problem Solver

June 2026 – Aug 2026

- Worked directly with the CEO to define product direction, prioritize engineering initiatives, and translate customer problems into production AI systems for an LLM-powered shopping assistant.
- Built a production evaluation platform analyzing **49,807 conversations** and **80,124 query-level routing decisions**, achieving **97.7% evaluation coverage** through versioned routing audit and historical backfill pipelines that reprocessed 47,231 conversations spanning 100+ days of production history.
- Developed end-to-end experimentation infrastructure across frontend, backend, workers, and admin systems, improving recommendation completion from **62.8%** to **75.0% (+12.2 percentage points)**.
- Evolved Shoppport's routing stack through three architectural generations – LLM classification, an embedding-based semantic fast path ($\sim 8\times$ faster for confident cases), and a hybrid classifier for ambiguous queries – reaching **86% V3 routing accuracy** and rescuing **93%** of eligible router errors.
- Designed and shipped Wowness, a Mem0-backed cross-category discovery recommender grounded in the live catalog, as sole engineer across a 919-line NestJS engine, React UI, and admin review lab – extending the Memory-first Personalization architecture from proposal to production.
- Diagnosed production failures at the system boundary, including a split MongoDB connection that held the V3 feature switch at zero executions, and corrected systematic analytics false positives, reducing one queue-level count from 2,005 to 811.
- Contributed **730 GitHub commits** and **143 merged pull requests**, adding over **114,000 lines** of TypeScript/TSX during the engagement.

SIGNUM (Nittany AI Alliance – PIT-UN Funded)

Aug 2024 – Nov 2025

Co-founder of a \$3,750 PIT-UN funded AI platform for caregiver decision support.

- Built multi-source ETL pipelines integrating CMS, NPPES, and Google Places datasets into a DuckDB warehouse.
- Developed hospital-rating prediction models using Markov Transition analysis with confidence intervals.
- Designed and deployed domain-specific RAG systems using FAISS and AWS Bedrock, reducing caregiver research time from 50+ hours per week to near-instant answers.

Penn State Advanced Vehicle Team (EcoCAR)

Project Manager, Vehicle Systems Engineering

University Park, PA

Sept 2026 – Present

- Built and maintained Smartsheet project schedules for four major vehicle-engineering deliverables, including CAV architecture, facilities and safety readiness, baseline vehicle testing, and propulsion architecture selection.
- Translated technical requirements and grading rubrics into structured work breakdown schedules with task owners, estimated hours, milestones, dependencies, and internal review deadlines.
- Coordinated with software, hardware, controls, safety, and testing teams to identify cross-functional dependencies, external approvals, facility constraints, and vehicle-access requirements.
- Developed strategic drivers, critical-path analyses, and project risk plans for the team's Integrated Year 1 Workplan.
- Monitored project status, identified incomplete or delayed work, and communicated priorities and schedule changes to technical leads and student leadership.
- Prepared technical-review presentations and project documentation to demonstrate progress, evidence of work, remaining activities, and readiness for upcoming milestones.

Atom Tech Solutions LTD

Software Engineering Intern

Berkeley, CA

May 2022 – Aug 2022

- Built secure authentication systems using SQL/JWT authentication and encrypted credential storage.
- Implemented automated credential generation and authenticated onboarding workflows.

Technical Projects

GladIEEEators – Penn State IEEE Battle Bots (1st Place)

Apr 2026

- Designed Shot & Chaser, a coordinated two-robot combat system, for the IEEE Student Chapter Battle Bots Competition, winning 1st place.
- Engineered a belt-driven AR500 vertical-spinner weapon system (334 J at 18,500 RPM, 200+ mph tip speed), validated via Fusion FEA above 4,000 N with a 2.09 minimum safety factor.
- Designed mixed-material armor (6061 aluminum, UHMW, TPU) and integrated drivetrain, radio control, and protected LiPo electronics across both robots within a \$976 total build budget.

XIIO – LinkedIn for Filmmakers

Apr 2026 – Present

- Built a creator networking platform featuring portfolios, video streaming, collaboration workflows, and school-based discovery using Next.js, Firebase, and Cloudflare Stream.
- Designed real-time messaging, creator profiles, media uploads, and production portfolio management.

ASME @ Penn State Website

Aug 2024 – Present

- Developed the official ASME Penn State website using React, TypeScript, Vite, and Firebase.
- Designed a multi-tier permission and approval system for organization management.

Hangukgwan (Family Business Web App)

2024 – Present

- Developed a multilingual full-stack restaurant web application using React, TypeScript, Node.js/Express, and MongoDB Atlas.
- Implemented JWT authentication, Google Maps API integration, and multilingual i18n support.

Teaching Experience

Lead Teaching Assistant, Math 230 (Multivariable Calculus) Penn State University Aug 2025 – Present

Responsibilities include assisting instruction, supporting students during office hours, and helping coordinate course activities.

Leadership and Service

THON Technology Committee Technology Captain (2027) Apr 2026 – Present

Association for Computing Machinery (ACM) Web & Design Director Apr 2026 – Present

Republic of Korea Air Force Squad Leader Sept 2022 – June 2024

Relevant Coursework

Computer Science

- CMPSC 131: Programming and Computation I
- CMPSC 132: Programming and Computation II
- CMPSC 221: Object Oriented Programming
- CMPSC 311: Introduction to Systems Programming
- CMPSC 360: Discrete Mathematics
- CMPSC 461: Programming Language Concepts
- CMPSC 464: Theory of Computation
- CMPSC 465: Data Structures and Algorithms

Computer Engineering

- CMPEN 270: Digital Design
- CMPEN 331: Computer Organization and Design

Awards and Honors

1st Place, ThinkNeuro LLC Project Competition Apr 2026

1st Place, IEEE Student Chapter Battle Bots Competition Apr 2026

PIT-UN Funding Award (\$3,750) Apr 2025

3rd Place, ACM Pathfinder Robotics Challenge Apr 2025

Republic of Korea Air Force Commendation Award Aug 2023

Technical Skills

Programming Languages: Python, C, C++, Java, JavaScript, TypeScript, SQL

Frameworks: React, Next.js, Node.js, NestJS, Express, Flask, Firebase

Machine Learning: PyTorch, TensorFlow, Flower, Scikit-learn, spaCy, SHAP, FAISS, Mem0, AWS Bedrock, OpenAI API

Data and DevOps: PostgreSQL, MongoDB, Redis, DuckDB, Docker, Kubernetes, Helm, GitHub Actions, Datadog, Git

Project Management: Smartsheet, Work Breakdown Structure, Gantt Charts, Critical Path Analysis, Risk Management, Requirements Analysis, Dependency Tracking, Technical Documentation, Microsoft Teams, SharePoint, Word, PowerPoint